

GhanaMediaWatch

Frontend Architecture

What the mobile client does with everything the server sends — and, more usefully, the work it already handles so the server does not have to. Written for someone who needs to understand the client without working on it.

Audience backend engineer **Pairs with** API Contract §1-§10 **Built today** Phase 1 of 8

Stack Expo SDK 57 • RN 0.86

READ THIS FIRST – STATUS

This document describes the client **as designed**. Only Phase 1 is built today. Every section is marked **BUILT** or **SPECIFIED** — **do not assume a Specified component exists yet**.

PHASE	SCOPE	STATUS
1	Scaffold, tooling, design system, tab shell	BUILT
2	Types, ApiClient , MockApiClient , API contract	CONTRACT ONLY
3	Local DB + sync engine	SPECIFIED
4	Capture flow — GPS gate, camera, review, outbox	SPECIFIED
5	Feed, map, incident detail	SPECIFIED
6	Auth and profile	SPECIFIED
7	Org / institution tier	SPECIFIED
8	Polish — states, a11y, i18n, performance	SPECIFIED

Division of labour

Several things a server might expect to handle are already solved on the device. Knowing which saves you building them twice.

CLIENT OWNS

- Media capture and the GPS gate
- Local persistence of media and metadata
- The outbox queue and its state machine
- Retry scheduling and exponential backoff
- Background upload continuation
- Connectivity detection, Wi-Fi-only deferral
- Chunk resumption bookkeeping
- Rendering of vetting state and privacy flags

SERVER OWNS

- Durable storage and media hosting
- Vetting decisions and routing
- The public feed and its ordering
- Map clustering
- Org saved queries and matching
- Authorisation, in full
- Abuse handling and lawful process

Both own display-flag enforcement. That is deliberate duplication, not redundancy — see §07.

CONCERN	CHOICE	VERSION	WHY NOT THE ALTERNATIVE
Framework	Expo, managed + dev client	SDK 57	Declarative native config while still allowing custom native modules.
Runtime	React Native, New Arch	0.86.2	React 19.2.3, Hermes.
Routing	Expo Router, typed	57.0.14	Route files stay thin; typed routes catch broken links at compile time.
Styling	NativeWind v4	4.2.6	One styling language app-wide. v5 is still preview.
Client state	Zustand	5.0.15	Capture session and upload queue are read from non-React code (the sync engine). Redux Toolkit is heavier than this needs.
Server state	TanStack Query	5.101.4	Caching, pagination and offline persistence <i>are</i> the problem; hand-rolling them is the mistake.
Local DB	expo-sqlite + Drizzle	0.45.2	The outbox is a relational queue with a state machine. AsyncStorage means hand-rolled indexing over JSON blobs.
Camera	vision-camera	5.2.2	Finer control over format, torch and frame quality than <code>expo-camera</code> .
Lists	FlashList	2.0.2	Feed and outbox must hold 60 fps on mid-range Android.
Forms	react-hook-form + zod	7.85 / 4.4	One schema drives both validation and the TS types on the submit payload.

TypeScript is strict, with `noUncheckedIndexedAccess` and `noUnusedLocals`. `any` is an ESLint *error*, not a warning.

Structure

BUILT

SRC/

TREE

app/	<i>expo-router routes only – thin, they compose features</i>
features/	
capture/	<i>GPS gate, camera, review, metadata editor</i>
feed/	<i>vertical pager</i>
outbox/	<i>offline queue UI + sync engine</i>
incident/	<i>detail view, map, navigation handoff</i>
org/	<i>institution / business tier</i>
auth/ settings/	
components/ui/	<i>design-system primitives</i>
api/	<i>ApiClient interface, MockApiClient, HttpApiClient</i>
db/	<i>drizzle schema, migrations, repositories</i>
services/	<i>location, permissions, device identity, sync, media</i>
stores/	<i>zustand stores</i>
hooks/ types/ lib/ i18n/	

Two rules hold this together: **route files contain no business logic**, and **cross-feature imports go through** `components/ui`, `services`, or `types` **only** — `features/feed` never imports from `features/capture`.

§ 04

API layer

SPECIFIED

Every screen is built against one swappable interface, with two implementations selected by a single environment variable.

.ENV

ENV

EXPO_PUBLIC_API_MODE=mock	→ <i>MockApiClient – seeded fixtures, artificial latency, injectable failure rate, no backend</i>
EXPO_PUBLIC_API_MODE=http	→ <i>HttpApiClient – real calls to API_URL</i>

This is why frontend work never blocks on the backend, and why the contract is already exercised end to end before a server exists.

WHAT THIS MEANS FOR YOU

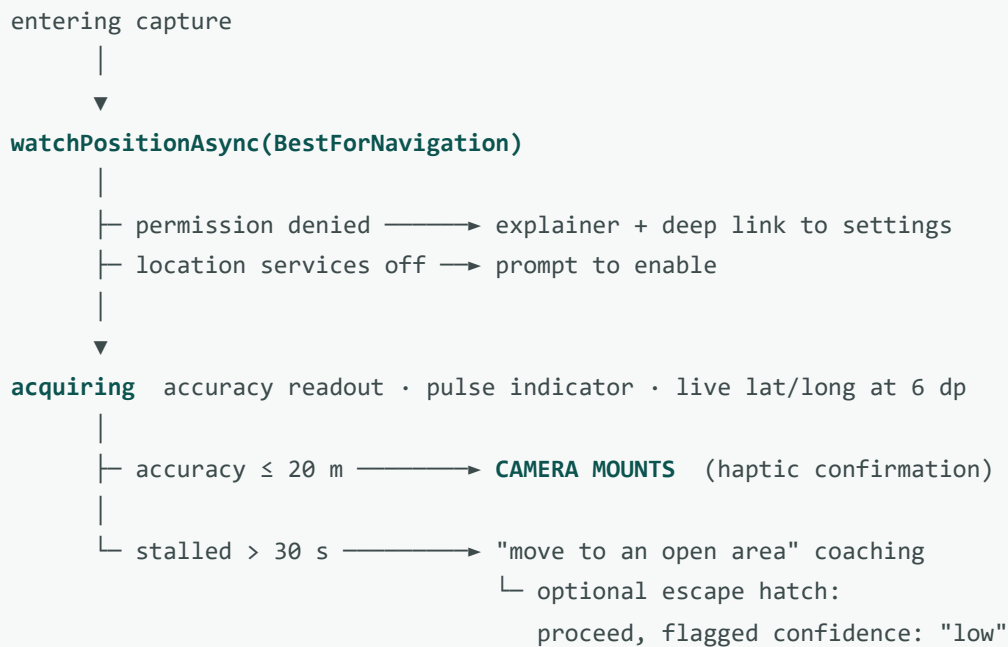
The client **never constructs a request inline in a component**. If your implementation needs to differ from the contract, the change lands in exactly one file — so raise differences early rather than working around them.

§ 05

The GPS gate

SPECIFIED

The defining feature. The camera preview does not mount until a location fix meets the accuracy threshold. This is a hard gate, not a warning.



CONSTANT	VALUE	RATIONALE
GPS_ACCURACY_THRESHOLD_M	20	About the best a consumer phone achieves outdoors within a few seconds. Tighter feels broken in urban canyons; looser is useless for sending someone to the scene.
GPS_STALL_TIMEOUT_MS	30 000	When coaching and the escape hatch appear.
GPS_ABSOLUTE_MAX_ACCURACY_M	150	Beyond this, not offered even via the escape hatch.
MAX_VIDEO_DURATION_S	60	Enforced client-side with a ring countdown.

The fix is locked at the moment of capture and does not keep updating afterwards.

Tamper signals sent alongside it: `accuracyM`, `altitude`, `heading`, `speed`, `capturedAtIso` (device clock), `capturedAtUtcOffsetMinutes`, `deviceUptimeMs`, and `isMocked` on Android. The reduced-accuracy escape hatch *flags* the report — it never silently downgrades it, and the flag is visible in the UI as well as the payload.

§ 06

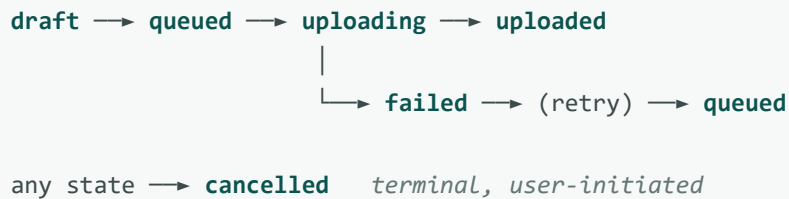
Offline sync

SPECIFIED

The network is treated as absent by default. Capture always succeeds locally.

On capture the client writes the media file to `documentDirectory/incidents/` — **never the cache directory**, which the OS can evict — and inserts a row in the local `incidents` table. Only when both succeed does the UI report success.

Local record state machine



Persisted per row: `attemptCount` , `lastError` , `nextRetryAt` , `serverId` , `bytesUploaded` .

Retry schedule

ATTEMPT	DELAY	NOTES
1	2 s	Plus $\pm 25\%$ jitter, so a thousand phones do not retry in lockstep when a cell tower returns. A <code>Retry-After</code> header on a <code>429</code> overrides the schedule.
2	8 s	
3	30 s	
4	2 min	
5	10 min	
6+	1 h	

Drain rules

- Drains when `isInternetReachable === true` — **not merely** `isConnected` , which is true on a captive-portal Wi-Fi that goes nowhere.
- Oldest first, one item at a time.
- A background task continues uploads when the app is backgrounded. **iOS background execution is opportunistic and not guaranteed** — the app also resumes draining on foreground. This is documented, not assumed away.
- **Wi-Fi-only toggle** defers *video* uploads to Wi-Fi; photos still go over mobile data. The outbox shows what is waiting and why.

CONSEQUENCE FOR RETENTION

The local media file is deleted **only after the server confirms completion**. The metadata row is kept, with `serverId`, so the reporter retains a personal history.

Which is why media retention is question 1 in the contract's open questions: **if the server expires footage the client has already deleted, the user's own history points at nothing.**

§ 07

Privacy enforcement

SPECIFIED

Three booleans per report: `showLocation`, `showDate`, `showTime`. **The client enforces these independently of the server**, with unit tests asserting a hidden value never reaches rendered output.

WHY BOTH LAYERS EXIST

Client-side hiding alone means true coordinates travel to every phone that scrolls past the report and sit in an inspectable cache. Server-side alone means one client bug leaks them. **Neither layer is the backup for the other.**

The review screen shows a **live public preview** — exactly what a stranger will see — so the reporter is never guessing what they are about to publish.

Anonymous mode

A prominent toggle with an honest explainer sheet. The copy states that anonymous means the public and other users see no profile, **but a device identifier is still attached** for abuse prevention and accountability. The client deliberately does not overstate the guarantee — and `deviceId` is never rendered in the public UI, on any screen.

Device identity

SPECIFIED

- A **UUID v4** generated on first launch, persisted in the secure keychain. Primary identifier, sent as `X-Device-Id` on every request.
- Supplemented by the Android ID / iOS `identifierForVendor` as a secondary field.
- **Both platform IDs reset on uninstall or factory reset and are not stable.** Treat them as a weak correlation signal; the keychain UUID is the primary.

Design system

BUILT

Tokens live as CSS custom properties, surfaced as Tailwind utilities. Components never hardcode a hex value.

The palette is tuned for a 70-year-old reader, which drove some non-obvious choices:

- **Light by default, not dark.** Intraocular light scatter roughly doubles between 20 and 70, so white-on-black haloes badly. Dark mode is a young-eye preference.
- **WCAG AAA (7:1) for body text**, not AA — contrast sensitivity falls sharply after 60.
- **Warm paper, not pure white** — `#FFFFFF` at phone brightness causes veiling glare.
- **Warm hues.** The aging lens yellows and absorbs short wavelengths, so blues muddy. Blue never carries meaning alone.
- **17px body type, 48pt tap targets.**

Category colours are grouped into families so hue signals the *kind* of incident and the label says which one. Colour is a secondary cue — twelve hues cannot be reliably

distinguished by an older eye, so every chip carries a text label and every map pin gets a distinct icon.

All of this is **enforced by 33 test assertions**, not just documented — a later "soften that grey" fails CI.

§ 10

Constraints

#	CONSTRAINT	CONSEQUENCE
1	vision-camera v5 ships no Expo config plugin	Camera and mic permissions declared directly in <code>app.json</code> . Re-check if the library restores one.
2	<code>react-dom</code> pinned to 19.2.3	<code>expo-router</code> → <code>@expo/ui</code> → <code>vaul</code> floats it past Expo's pinned React and breaks npm peer resolution.
3	Android cannot synthesise font weights	Each Inter weight is registered as its own family: <code>font-sans</code> , <code>-medium</code> , <code>-semibold</code> , <code>-bold</code> .
4	Testing Library v14 made render async	Every test must <code>await</code> render and <code>fireEvent</code> .
5	Emulators report a perfect mocked fix	Precisely the branch the gate detects. Capture needs a physical device to test meaningfully.

§ 11

Takeaways

If you read nothing else in this document, read these five.

BECAUSE THE CLIENT...	YOU SHOULD...
is offline-first and retries forever	Assume duplicates. Implement idempotency keys — this is not optional.
already backs off on 429 and 5xx	Return the correct status and it behaves. Return <code>500</code> for a validation error and every affected phone retries hourly, indefinitely.
deletes local media on upload completion	Treat retention as a product decision, not an internal detail — it changes what a reporter sees in their own history.
enforces display flags too	Still suppress at source. That is defence in depth, not permission to send suppressed values.
is built against one swappable interface	Raise contract differences now. Cheap today, expensive once the screens exist.